

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. OVERVIEW OF FUNCTIONAL FEATURES

ComicCraft is packed with robust features designed to ensure an end-to-end automated user experience. The application bridges the gap between text-based ideation and formatted visual export.

3. DETAILED LIST OF FUNCTIONAL FEATURES

- **Feature 1: Interactive Web Input Form**
 - *Description:* A clean, responsive HTML/CSS form allowing users to input customized parameters including Story Prompt, Character Name, Setting, Tone (e.g., Dramatic, Funny), and Art Style (e.g., Anime, Realistic, Comic Book).
- **Feature 2: Dual-Model LLM Story Orchestration**
 - *Description:* Combines Google Gemini 1.5 Flash (for rapid structured JSON outline generation) and Gemini 1.5 Pro (for rich, creative dialogue and narrative expansion) to build compelling comic storylines.
- **Feature 3: Multi-Engine AI Image Generation Pipeline**
 - *Description:* Automatically translates panel scene descriptions into high-quality illustration prompts, routing queries through Pollinations AI nodes and Hugging Face Stable Diffusion models.
- **Feature 4: Advanced Exception Handling & API Fallbacks**
 - *Description:* Implements automated error detection for HTTP 402, 500, and 503 status codes, instantly switching between API routes or defaulting to a custom Python PIL-rendered "Comic Caption Panel" to prevent application failure.
- **Feature 5: Dynamic Web Preview Rendering**
 - *Description:* Utilizes Jinja2 templating loops to render the generated comic panels sequentially on an interactive preview page

(comic_preview.html), displaying panel titles, illustrations, and narrations cleanly.

- **Feature 6: Automated PDF Compiler and Exporter**

- *Description:* Integrates the FPDF library to dynamically calculate layout coordinates, bind generated images with corresponding text, and compile a professional, downloadable multi-page PDF document (comic.pdf).

- **Feature 7: Secure Environment & Dependency Management**

- *Description:* Uses python-dotenv for secure runtime injection of API keys and isolates project packages using a dedicated Python virtual environment and requirements.txt.